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ETE 3351
Lab 11 and Extra Credit

DIGITAL SIGNALS PART 1: PULSE WIDTH (PWM) MODULATION SIGNALS LAB

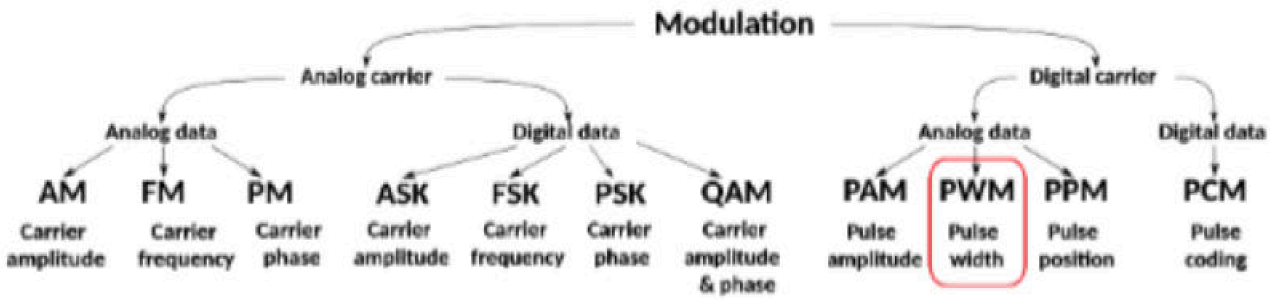
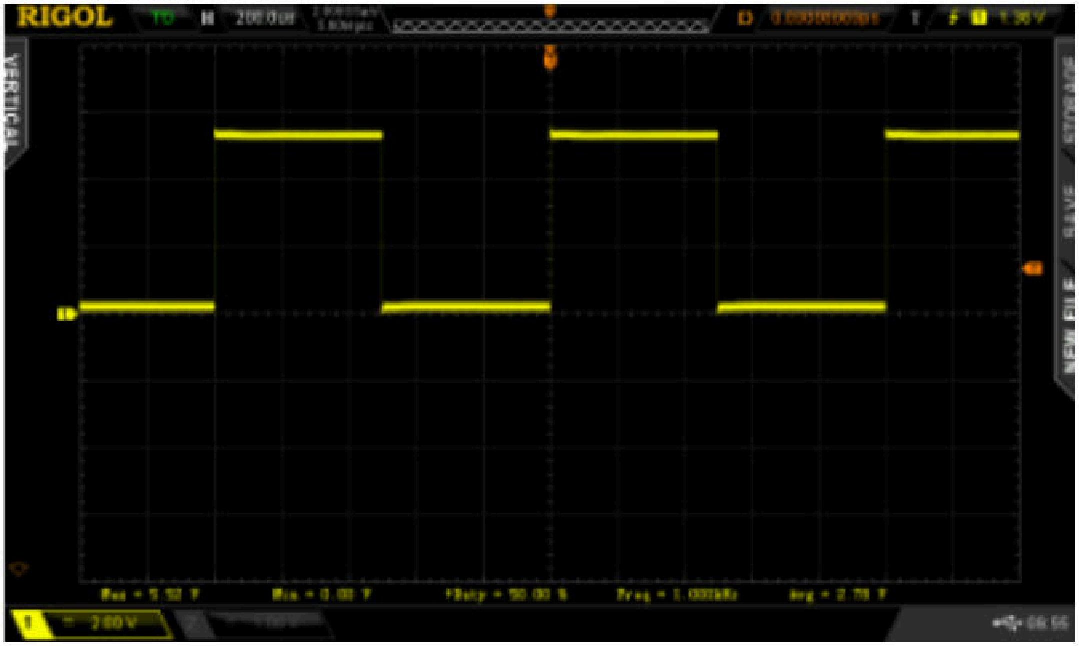


Figure from [Signal modulation - Wikipedia]

Introduction

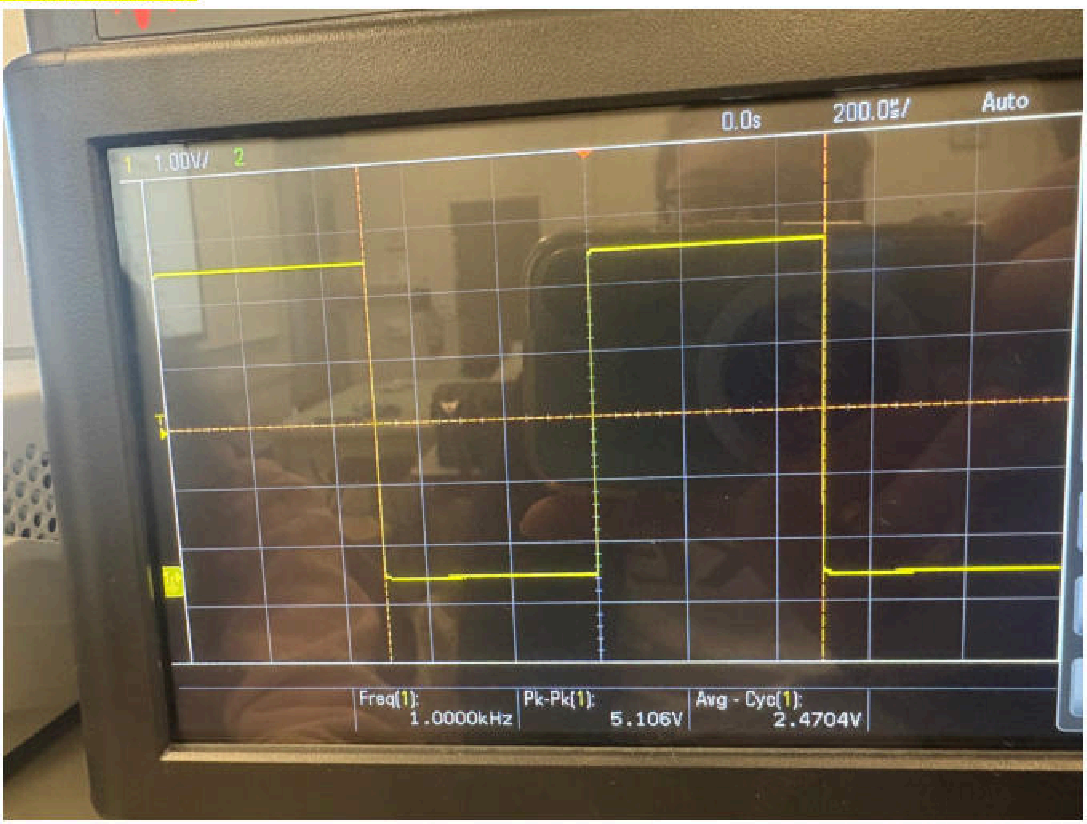
The objective of this lab is to view basic digital modulation signals on an oscilloscope and to examine Pulse Width Modulation (PWM) by relating the duty cycle of a digital signal to its average voltage. PWM is a digital modulation technique in which the signal switches between a low and high voltage level, and the information is carried by how long the signal stays high during each cycle, known as the duty cycle. Even though PWM is not commonly used in modern communication systems, it is widely used in practical applications such as motor control and RC servo motor control. In RC servo systems PWM is often implemented as a hybrid technique that combines traditional PWM with Pulse Position Modulation (PPM), as discussed during the lab. By adjusting the duty cycle PWM can represent analog information such as voltage level or signal shape. In this lab, PWM signals are generated and observed on a scope, and low-pass filtering is used to show how an analog signal can be recovered from a digital PWM waveform.

Setup 1: On the Rigol ARB generator select: PULSE, Default settings: 1KHz, 5Volt-pp no offset, Duty Cycle default ~50%



IMG 1. *expected*

No modulation:



IMG 2.

$V_{average} = (D * V_{high}) + ((1-D)\ V_{low})$

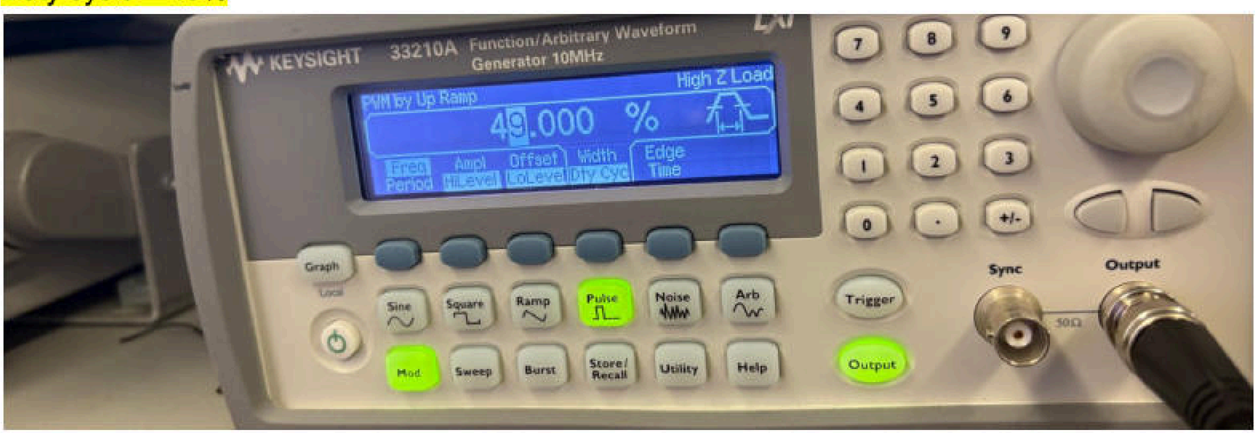
$2.4704 = D\ (5.1v) + (1-D)\ (0v)$

$D = 2.47 / 5.1 = 0.484 \approx 48.4\%$

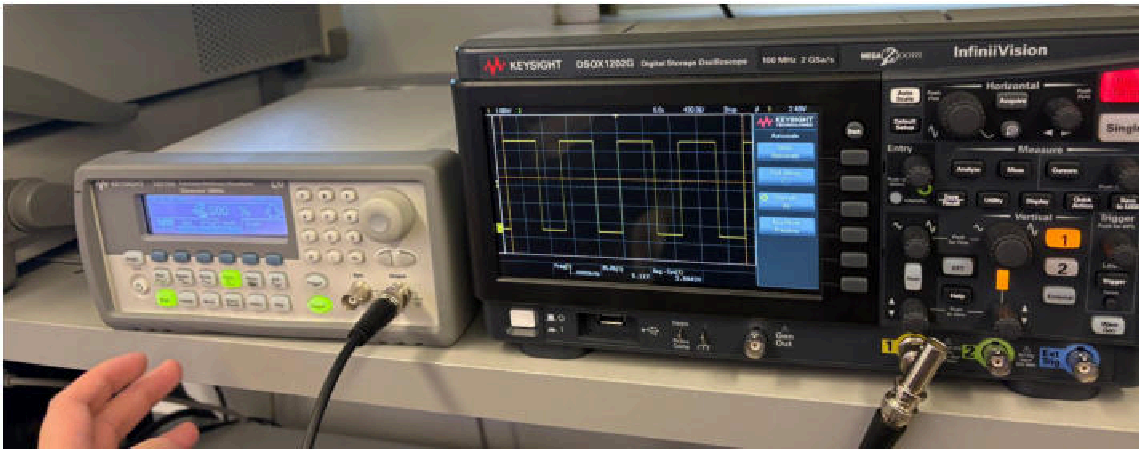
It can be seen that the default is not precisely a duty cycle of 50% and is instead approximately 48.4%. This tracks, as a duty cycle of 50% was not possible on the function generator in part 2 as well.

Step 2: Activate modulation MOD, if you see AM, FM... go to the 2nd screen using the small right hand red arrow button, near bottom right of screen to see PWM and IQ, see sample control.Select PWM and Set PWM Freq to 1Hz, Shape as UpRamp, and DutyCycle Dev (DutyDev) to 49% to account for low to high transition time. Review PWM signal generation, record various duty cycle values and their Average voltages: down to low % values (under 20%) and high % value >80%, use Single sweep button to freeze screen if needed. Beyond taking screenshots if you choose to you can make a plot of Duty Cycle and Vavg which should be a fairly straight line plot.

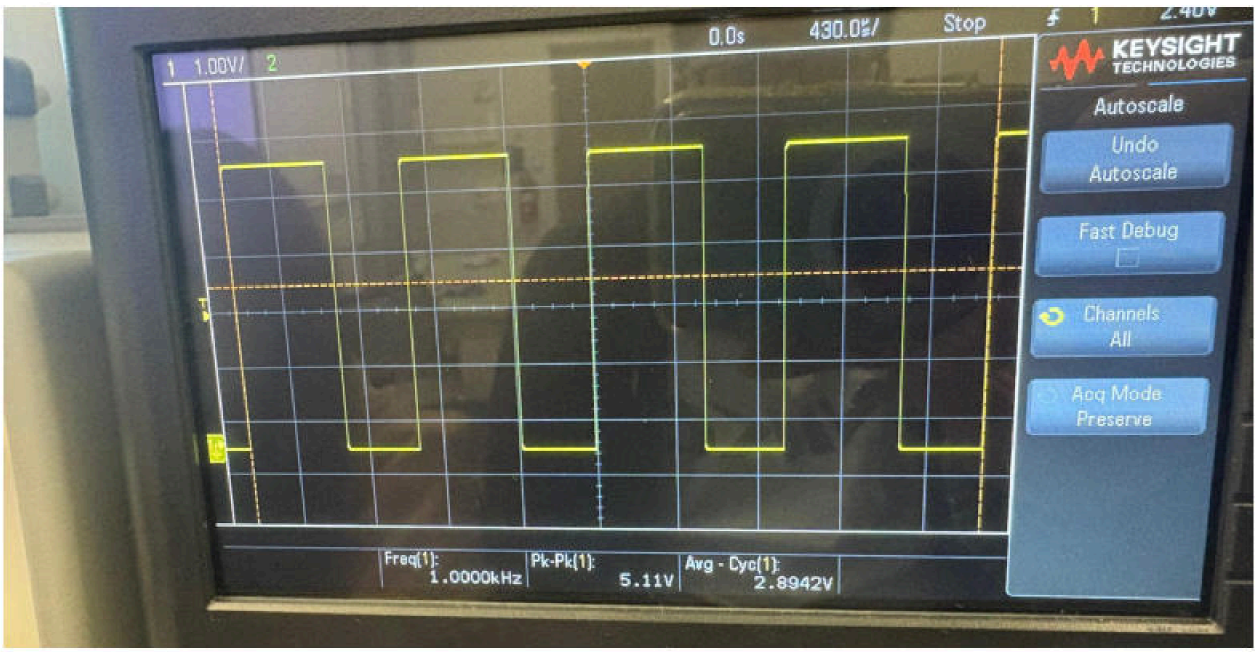
Duty cycle = 49%



IMG 3.

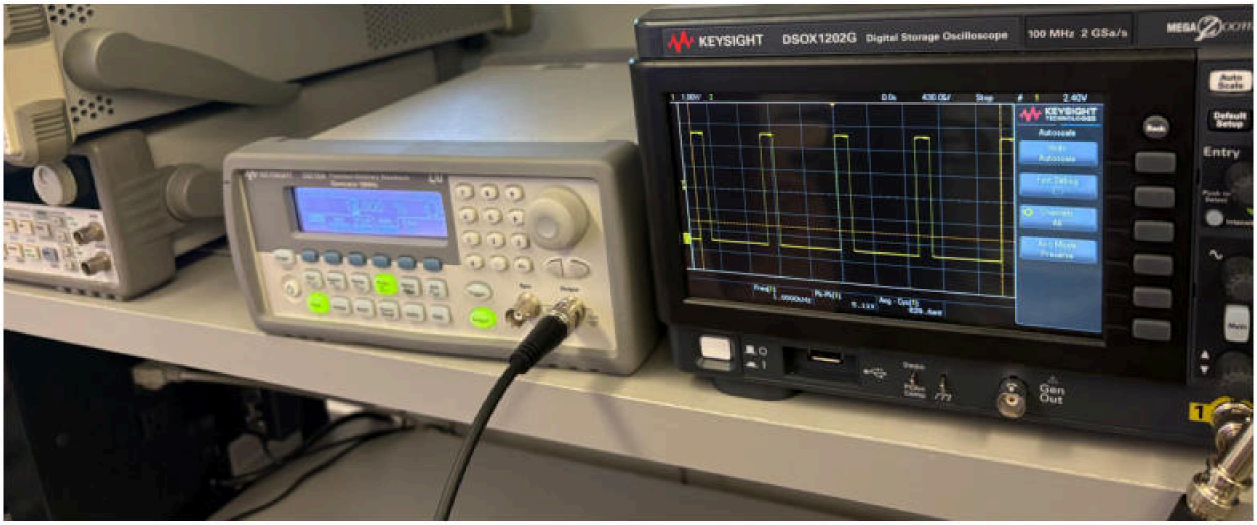


IMG 4.

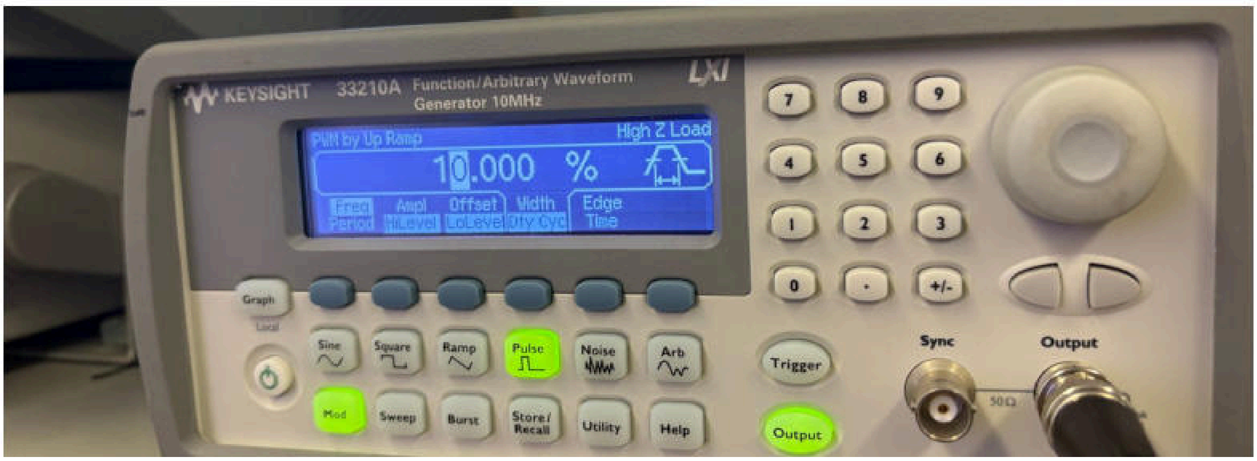


IMG 5.

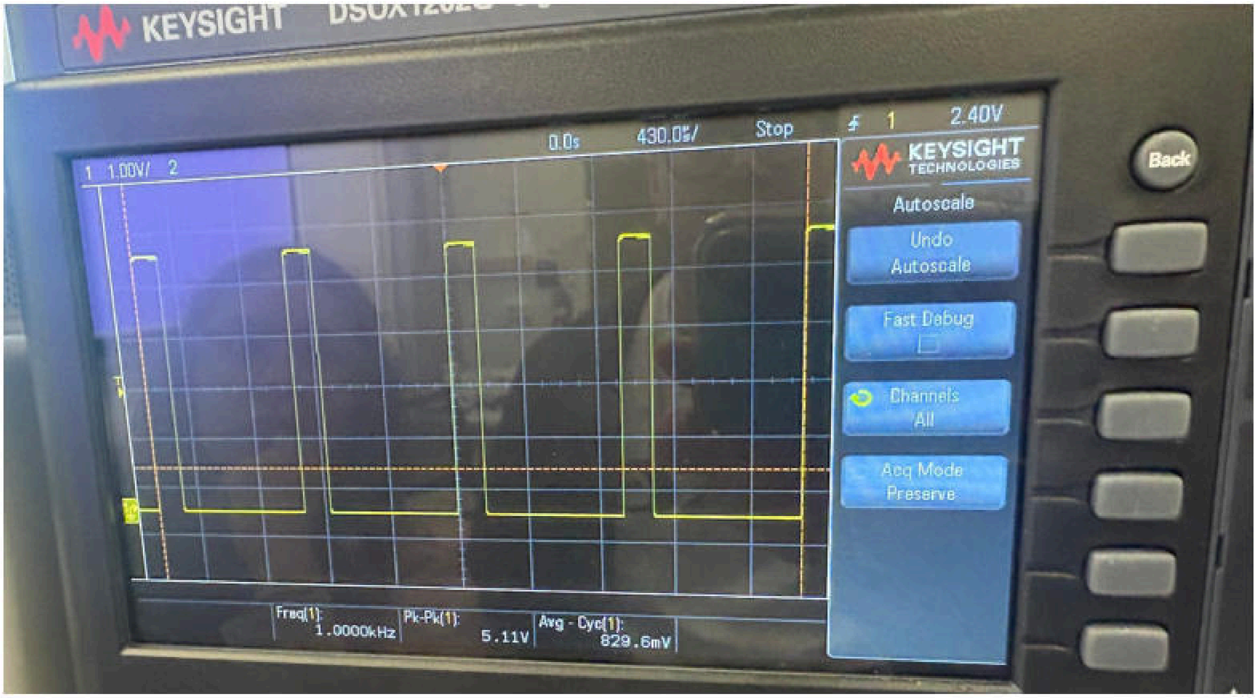
Under 20% (10% duty cycle)


KEYSIGHT
DSOX1202G Digital Storage Oscilloscope
100 MHz 2 GSa/s
Entry
Gen Out

IMG 6.


KEYSIGHT
33210A Function/Arbitrary Waveform Generator 10MHz
LXI
PWM by Up Ramp
High Z Load
10.000 %
Freq Period
Ampl HiLevel
Offset LoLevel
Width Dty Cyc
Edge Time
Graph
Sine
Square
Ramp
Pulse
Noise
Arb
Mod
Sweep
Burst
Store/ Recall
Utility
Help
Trigger
Output
Sync
Output
50 Ω

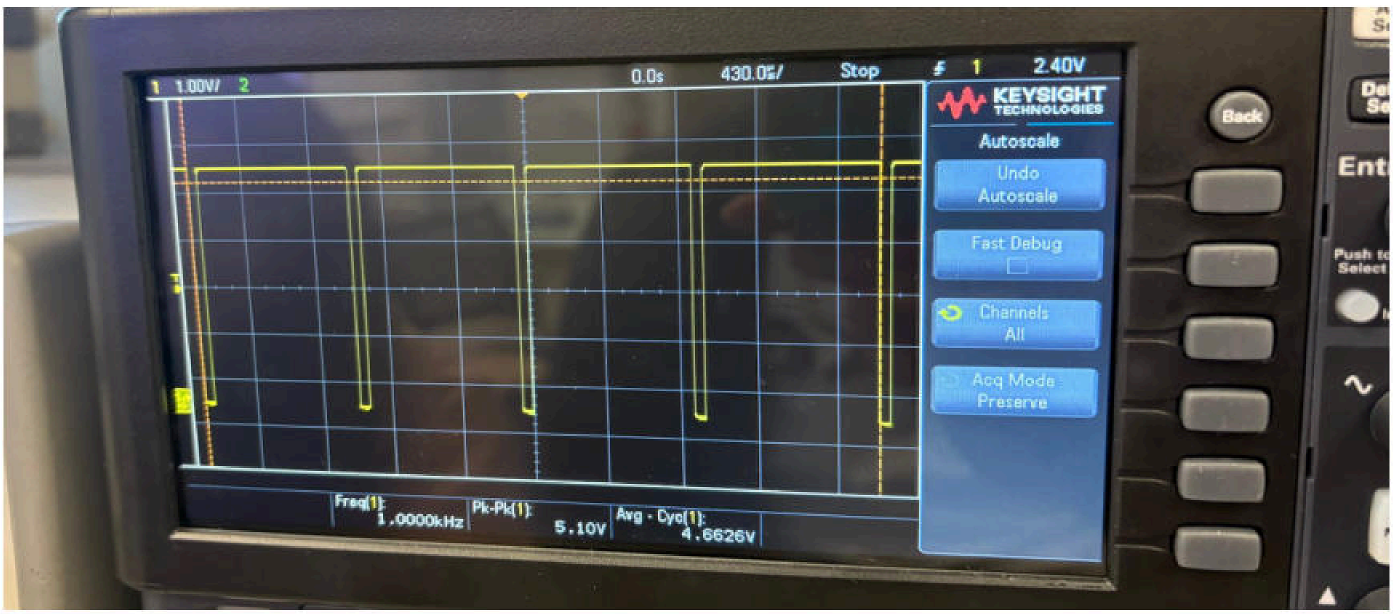
IMG 7.



IMG 8.

90% duty cycle

IMG 9.



IMG 10.

One of the most common PWM signals is sinusoidal PWM, where the pulse width or duty cycle is changed or modulated by a sinusoidal signal, as shown below. This waveform can look somewhat like FM modulation, often described as a "slinky effect," although it is closer to phase modulation (PM) since the phase is changing while the pulse period or frequency remains constant. The internal sinusoidal voltage modulates the pulse width, causing the signal to transition smoothly between narrower and wider pulses. In DC motor control applications, the PWM modulation percentage controls the average voltage (Vavg) applied to the motor and therefore controls the motor speed. The appropriate PWM frequency depends on the motor and other system factors, as discussed in the Electroboom video (https://youtu.be/yO9xlVv8ryc, time mark 12:30). The resistive–inductive (RL) low-pass nature of a motor filters out the high-frequency components of the PWM signal. Because of the sinusoidal nature embedded in this signal, sinusoidal PWM can also be used to run AC-type motors using variable speed or variable frequency drives, where the frequency is based on the PWM frequency and the voltage is controlled by the modulation signal. For the setup of sinusoidal PWM, the digital pulse is first reset to a 50% duty cycle with a frequency of 10 kHz. PWM modulation is then enabled, the modulation frequency is set to one-tenth of the pulse frequency (1 kHz), and the shape and duty cycle deviation (DutyDev) are set to 49%. Setting the duty cycle deviation to 49% for a 50% pulse wave causes the duty cycle to vary between 99% (50 + 49) and 1% (50 − 49). Single-sweep mode on the oscilloscope may be required to observe the full range of duty cycle variation

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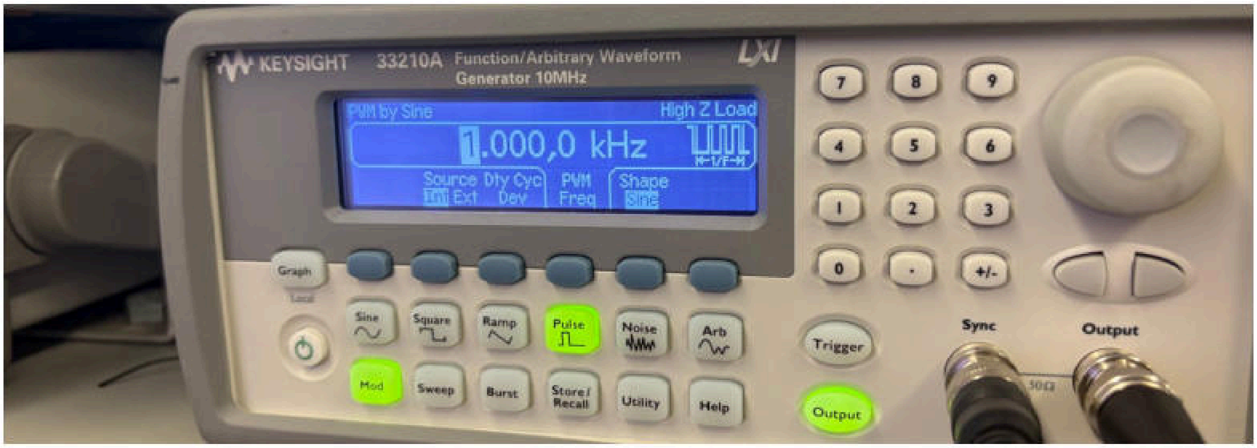
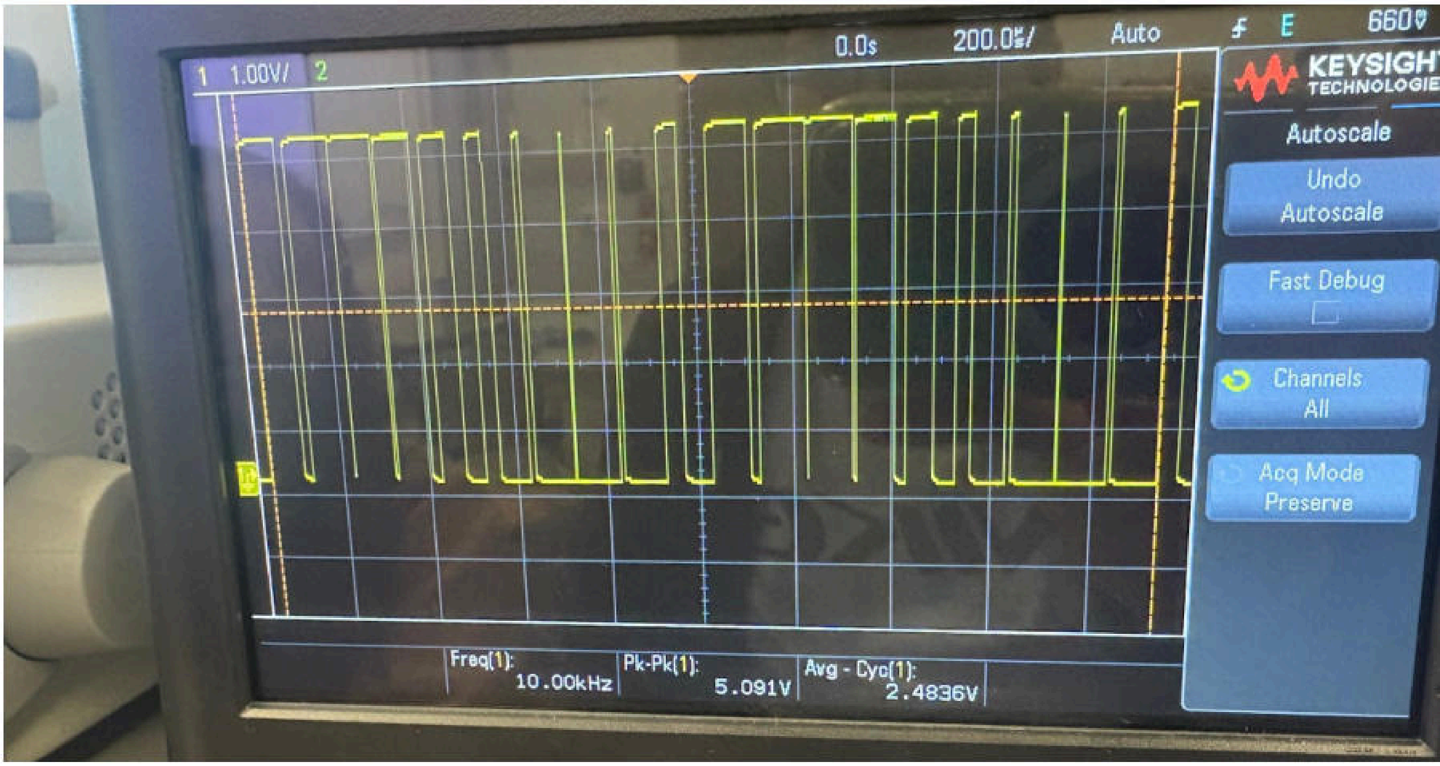
IMG 11. *Settings*

IMG 12. *Expected*



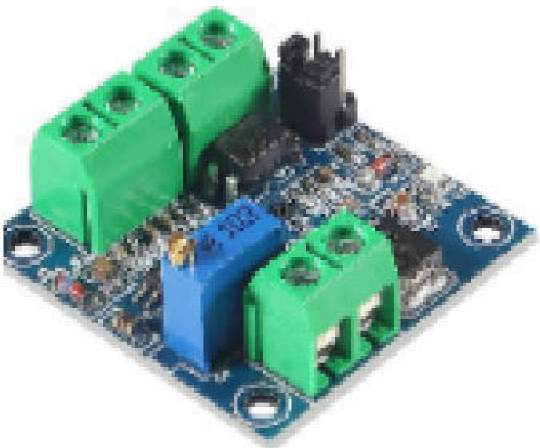
IMG 13. *Pulse frequency*



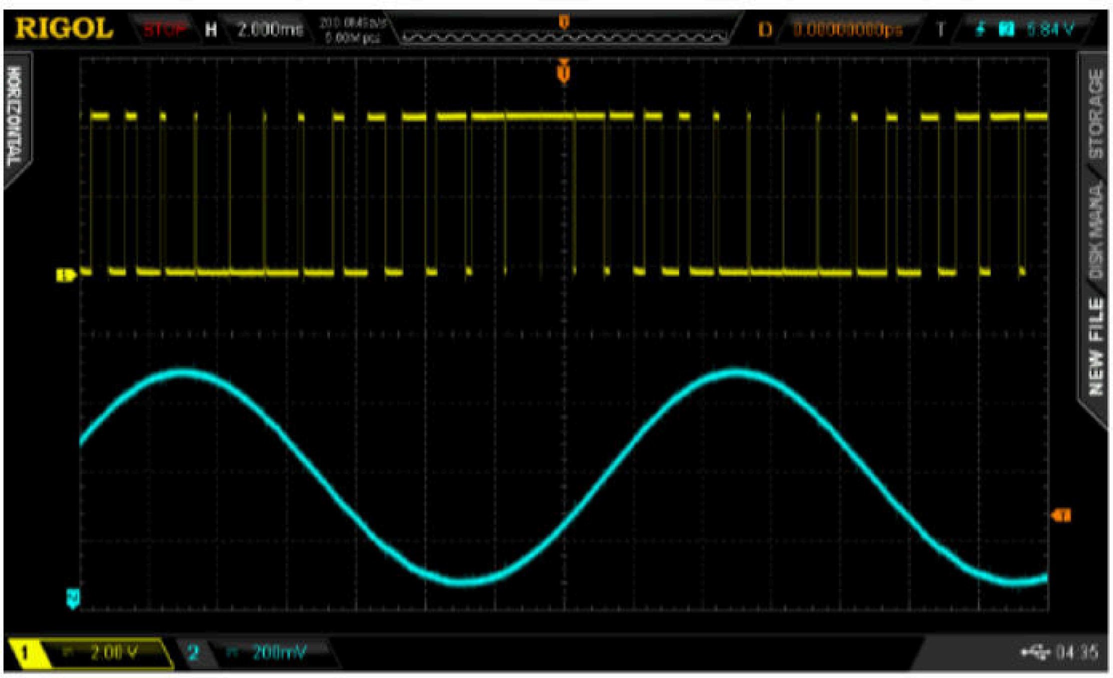
IMG 14. *Results*

Optional

As one can see in the screen picture the Duty Cycle is continuously changing from near 0%, a very narrow pulse to near 100% or a very wide pulse.Using a separate PWM to voltage converter circuit board (bought on Amazon and not required for lab) as a Demodulator we see an example of a sine wave PWM signal being demodulated, there is some phase delay between the two signals but the demodulated sine wave is clear



IMG 15.



IMG 16.

When the sine wave modulating the PWM frequency doubles the output frequency doubles as it should.

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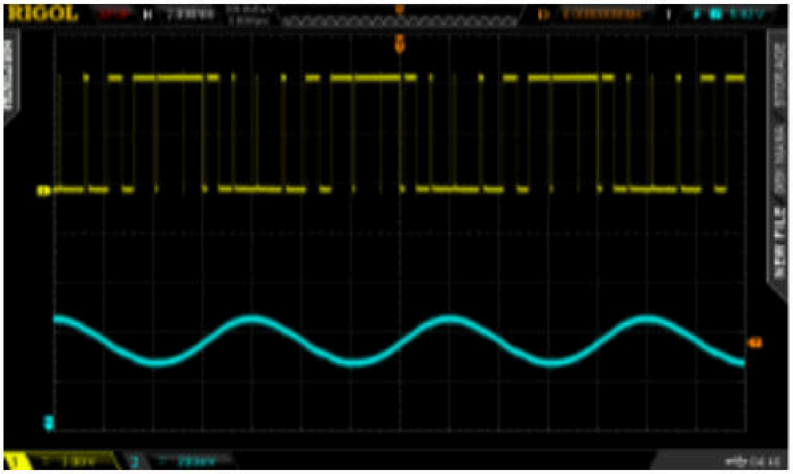
Extra Credit:

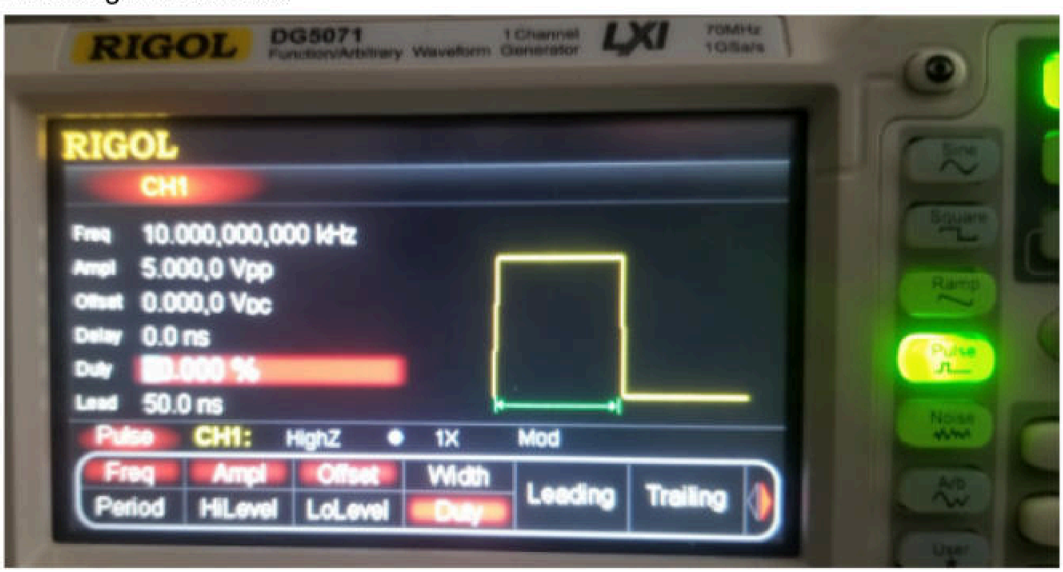
Use the scope cursors or time functions to measure the pulse widths from minimum to maximum and you should see the sinusoidal change in the pulse variation.



IMG 17

Addendum:

Based upon a suggestion from a student the PMW signal was also demodulated signal using a low pass filter since it also responds to the average value of the signal. Here are those results, for a 10kHz pulse signal 50% Duty Cycle and a 18kHz LPFilter. Sine wave demodulation was good, triangle and square not a good. Demodulated frequency was off but that is likely a setting in the Rigol Mod menu



IMG 18

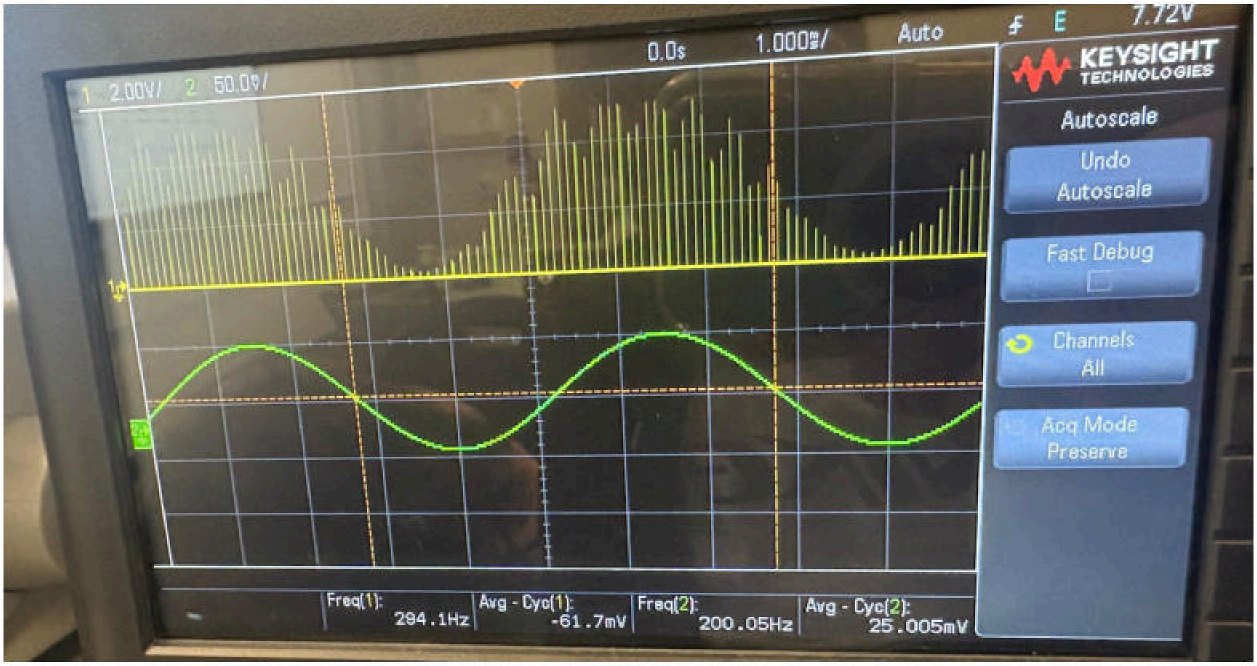
Sinewave modulation of 49%, so PWM goes from 50+49=99% to 50-49=1%


RIGOL
DG5071
Function/Arbitrary Waveform Generator
1 Channel
70MHz
1GSa/s
RIGOL
CH1
Type PWM
Source Internal
MFreq 200.000,000 Hz
Shape Sine
Duty 49.000 %
Mod CH1: HighZ 1X Mod
Type
Source Int
PWM Freq
Shape
DutyDev
Mod
Sweep
Burst
Sine
Square
Ramp
Pulse
Noise
Arb

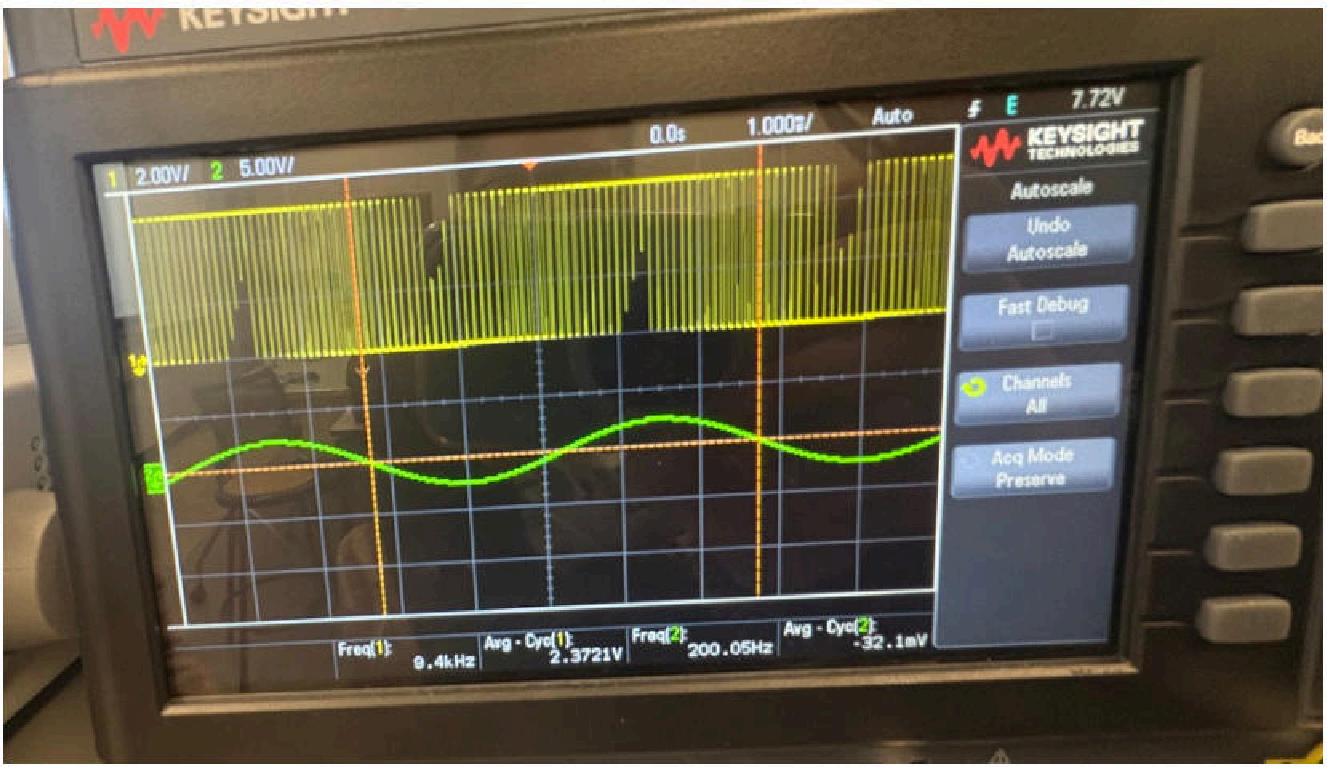
IMG 19


KEYSIGHT
33210A Function/Arbitrary Waveform Generator 10MHz
PWM by Sine
High Z Load
1.00 %
Graph
Sine
Square
Ramp
Pulse
Noise
Arb
Mod
Sweep
Burst
Store/ Recall
Utility
Help
Trigger
Output
Sync
Output

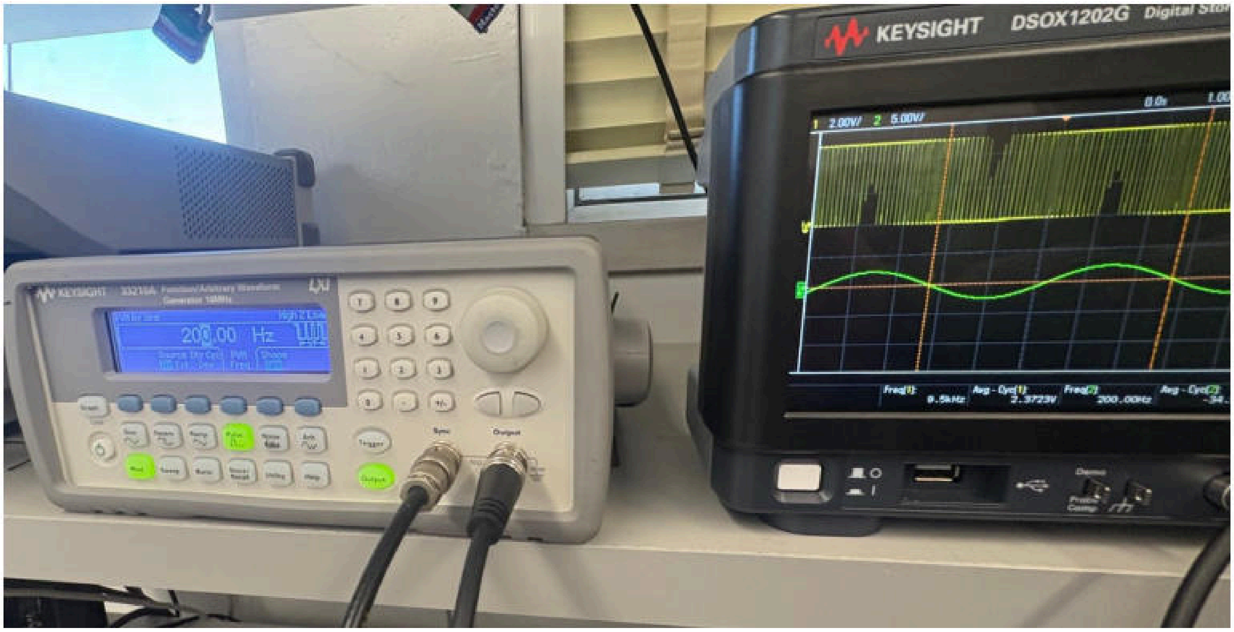
IMG 20


Auto
7.72V
KEYSIGHT
TECHNOLOGIES
Autoscale
Undo
Autoscale
Fast Debug
Channels
All
Acq Mode
Preserve
Freq(1): 294.1Hz
Avg - Cyc(1): -61.7mV
Freq(2): 200.05Hz
Avg - Cyc(2): 25.005mV

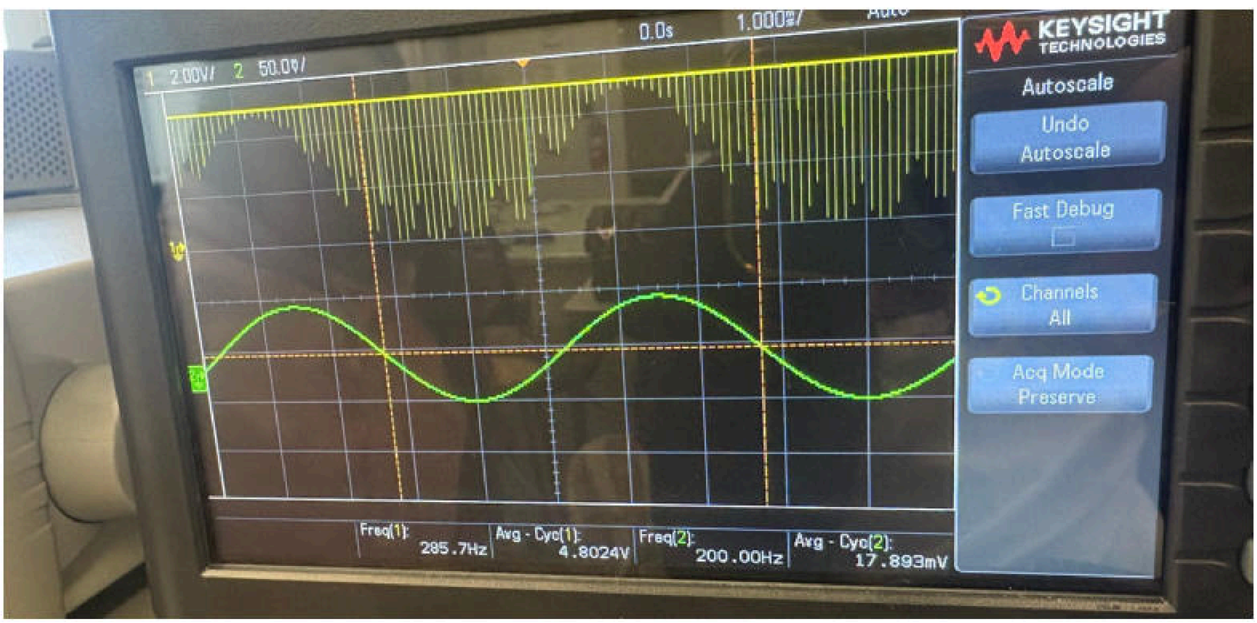
IMG 21


Auto
7.72V
KEYSIGHT
TECHNOLOGIES
Autoscale
Undo
Autoscale
Fast Debug
Channels
All
Acq Mode
Preserve
Freq(1): 9.4kHz
Avg - Cyc(1): 2.3721V
Freq(2): 200.05Hz
Avg - Cyc(2): -32.1mV

IMG 22.


KEYSIGHT
DSOX1202G
Freq(2): 200.00Hz

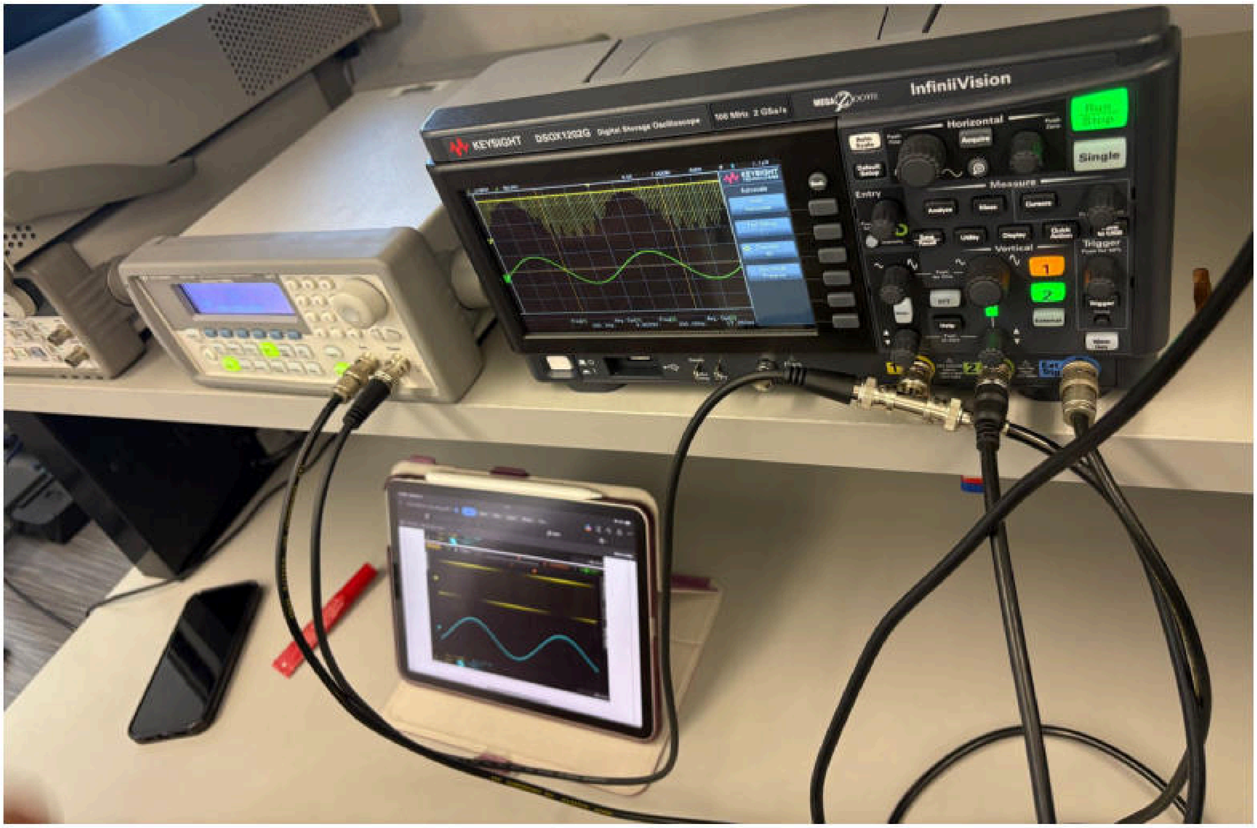
IMG 23


0.0s
KEYSIGHT TECHNOLOGIES
Autoscale
Undo Autoscale
Fast Debug
Channels All
Acq Mode Preserve
Freq(1): 285.7Hz
Avg - Cyc(1): 4.8024V
Freq(2): 200.00Hz
Avg - Cyc(2): 17.893mV

IMG 24.

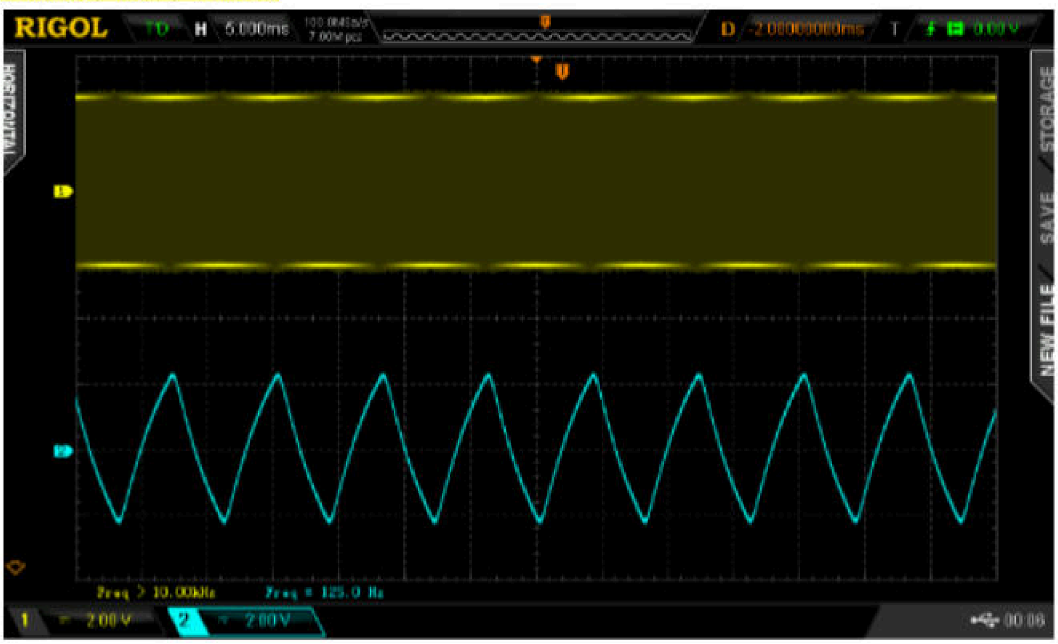
IMG 25

IMG 26. *Filter settings, right control is 18kHz LPF*

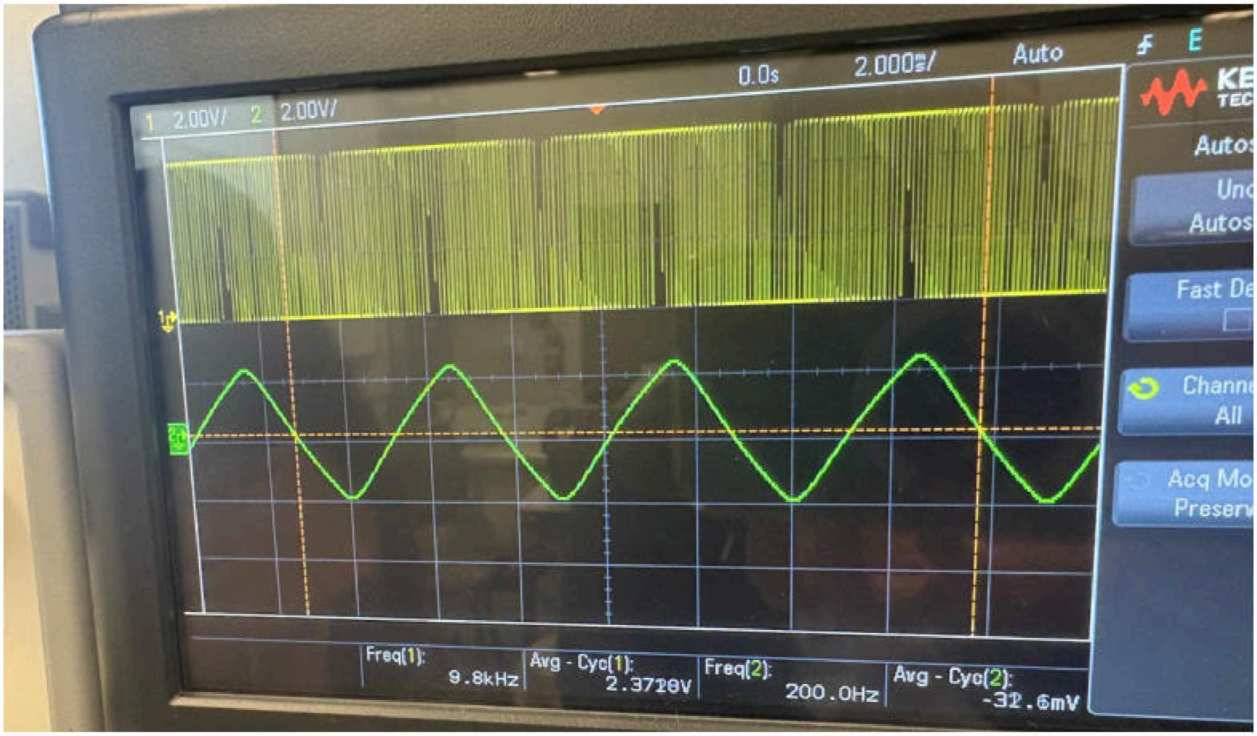


IMG 27. *System set up*

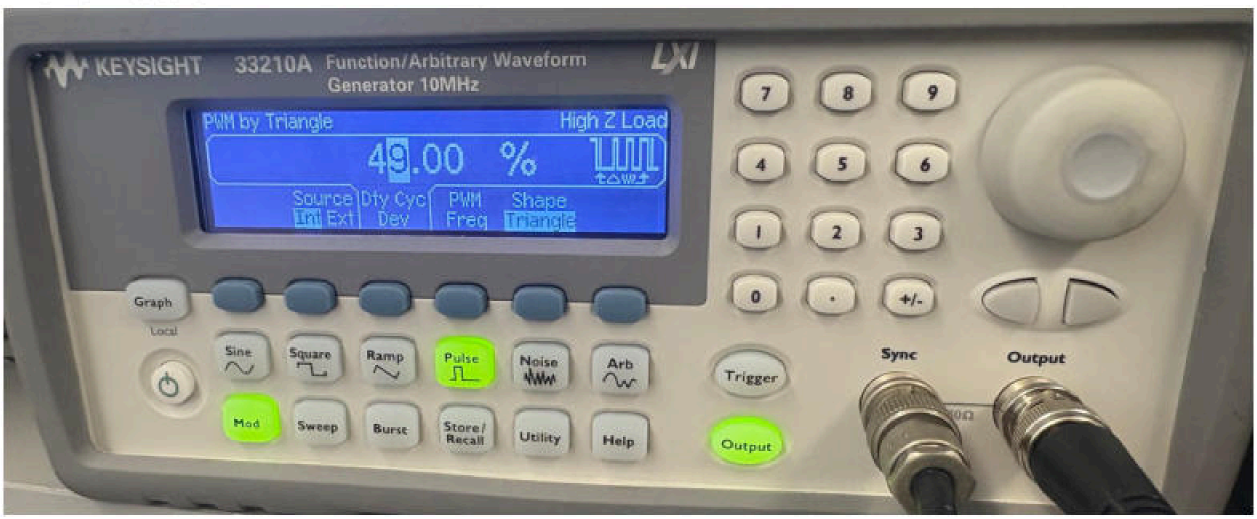
Triangle wave demod



IMG 28. *Expected results*

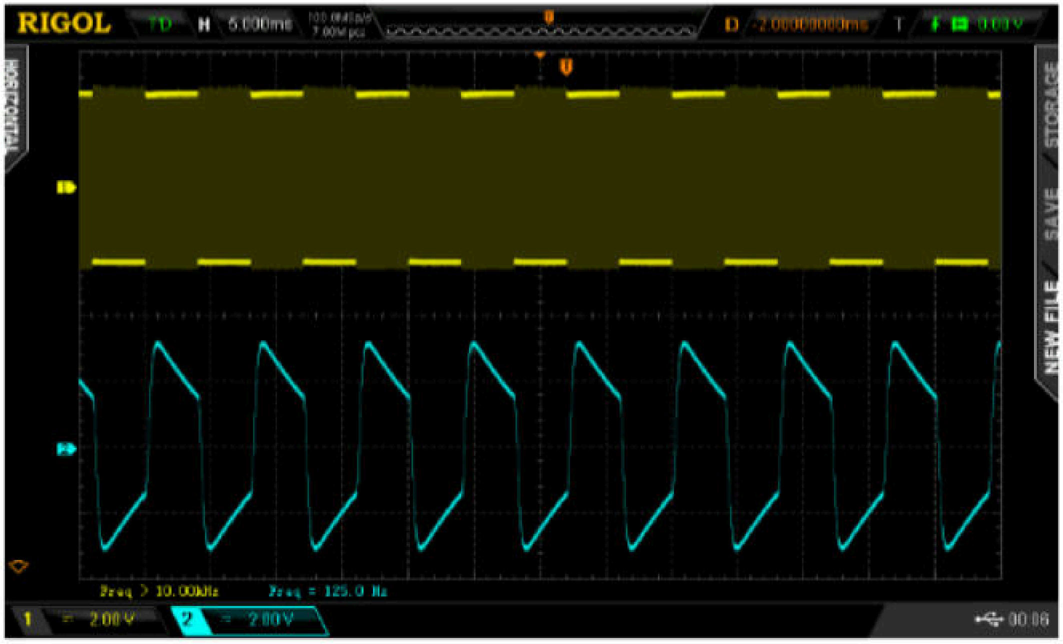


IMG 29. *Results*

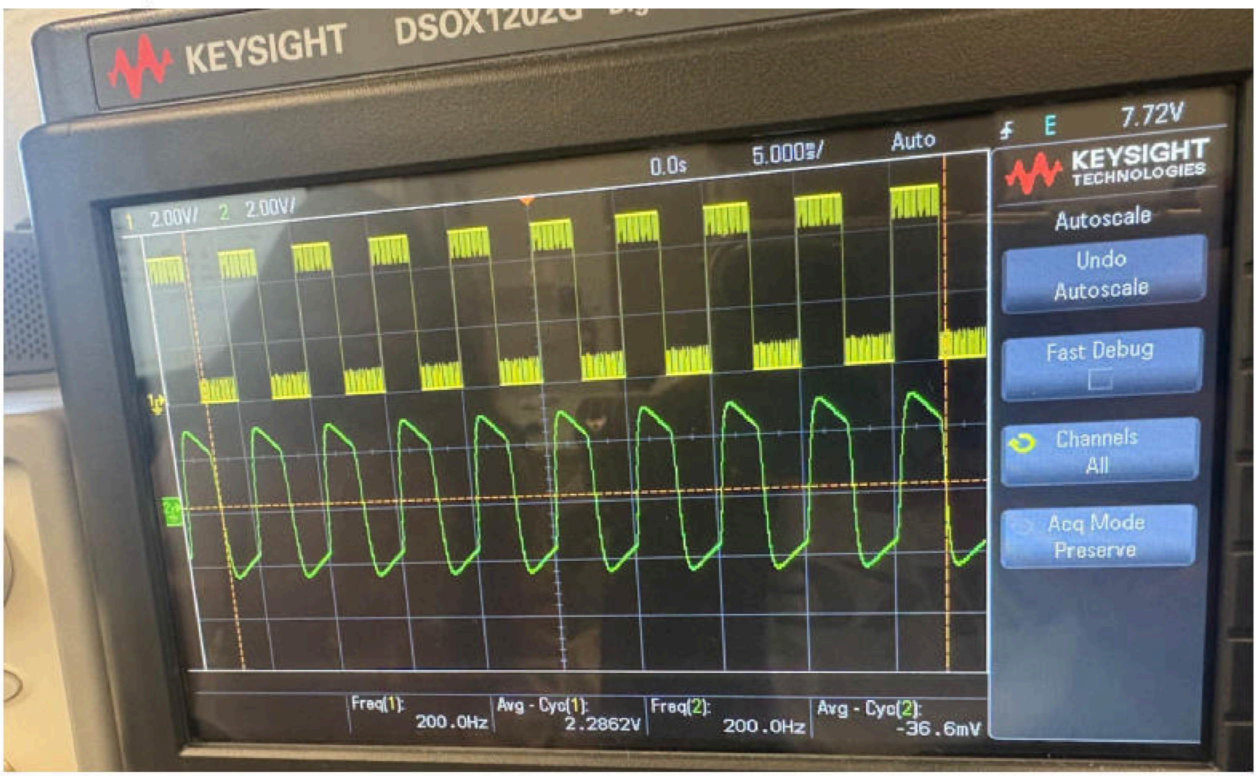


IMG 30.

Square wave demod



IMG 31. *Expected results*



IMG 32. *Our results*

IMG 33.

Conclusion

In this lab, we learned how Pulse Width Modulation (PWM) works and how it can represent analog information using a digital signal. By changing the duty cycle of a PWM signal we saw that the average voltage changes which explains how PWM is used to control things like motor speed (as discussed in the lab). We also observed sinusoidal PWM, where a sine wave smoothly changes the pulse width, and learned that low-pass filtering can recover a smooth waveform from the pulses. Overall, this lab showed how PWM controls voltage and power in a simple and efficient way.